① copy only
166/1218

CSU SCADA EQUIPPED

TENAGA NASIONAL BERHAD
Reviewed/Verified Engineer : Jdul Date : 23/10/17

IMPORTANT

Verification for the comformance to scheme and functionality. This verification does not relieve the contractor from full compliance to contractual obligation & technical specification

Pengesahan skim dan fungsi ini tidak melepaskan kontraktor daripada pematuhan kehendak kontrak & spesifikasi teknikal

- Final approval of the detailed schematic diagrams will be subjected to successful pre-commissioning test at site
- Approval is for secondary / primary equipments only
- There shall be no cost implication with this approval

EARTH FAULT INDICATOR

- pembetulan pd sheet 8

TECHNICAL SPECIFICATION

SCADA EQUIPPED FACILITIES FOR TNB COMPACT SUBSTATION 11/0.433kV

AMSDS
20 OCT 2017
TNB

MATERIAL	Type : -		COATING	Type : -	GENERAL TOLERANCE	PROJECTION
	Code : -			Code : -	X.X-XXXX ±5.0 UNLESS SPECIFIED	

Rev.	Reason	X	Y	Day	Month	Year
00	INITIAL RELEASE			20	OCT	2017

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : COVER PAGE

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABDUL MANAN

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No : CSU-SCA-EFI
Sheet 1 / 9

IM Code : -
Tender No. : -

Dept.	Year	Scale	Units
ENG	2017	-	mm

FORM NO: QA-F-01-12F

No.	DRAWING No.	SHEET No.	REV.	DESCRIPTION
1	CSU-SCA-EFI	1	00	COVER PAGE
2	CSU-SCA-EFI	2	00	DRAWING LIST
3	CSU-SCA-EFI	3	00	DIMENSION EMG EFI EASI-R
4	CSU-SCA-EFI	4	00	CT CONNECTION
5	CSU-SCA-EFI	5	00	VOLTAGE INPUT CONNECTION (OPTION)
6	CSU-SCA-EFI	6	00	SCADA CONNECTION 1 (OPTION 1)
7	CSU-SCA-EFI	7	00	EXTERNAL LED (OPTION)
8	CSU-SCA-EFI	8	00	TERMINAL FUNCTION DIAGRAM
9	CSU-SCA-EFI	9	00	DIP SWITCH SETTING

This document must not be copied without our written permission,and the contents there of must not be imported to a third paty not be used for any unauthorised purpose. Contravention will be prosecuted. INDKOM

AMSDS
20 OCT 2017
TNB

00	INITIAL RELEASE			20	OCT	2017	
Rev.	Reason	X	Y	Day	Month	Year	

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : DRAWING LIST

MATERIAL	Type : - Code : -	COATING	Type : - Code : -	GENERAL TOLERANCE X X-XXXX ±5.0 UNLESS SPECIFIED	PROJECTION

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABDUL MANAN

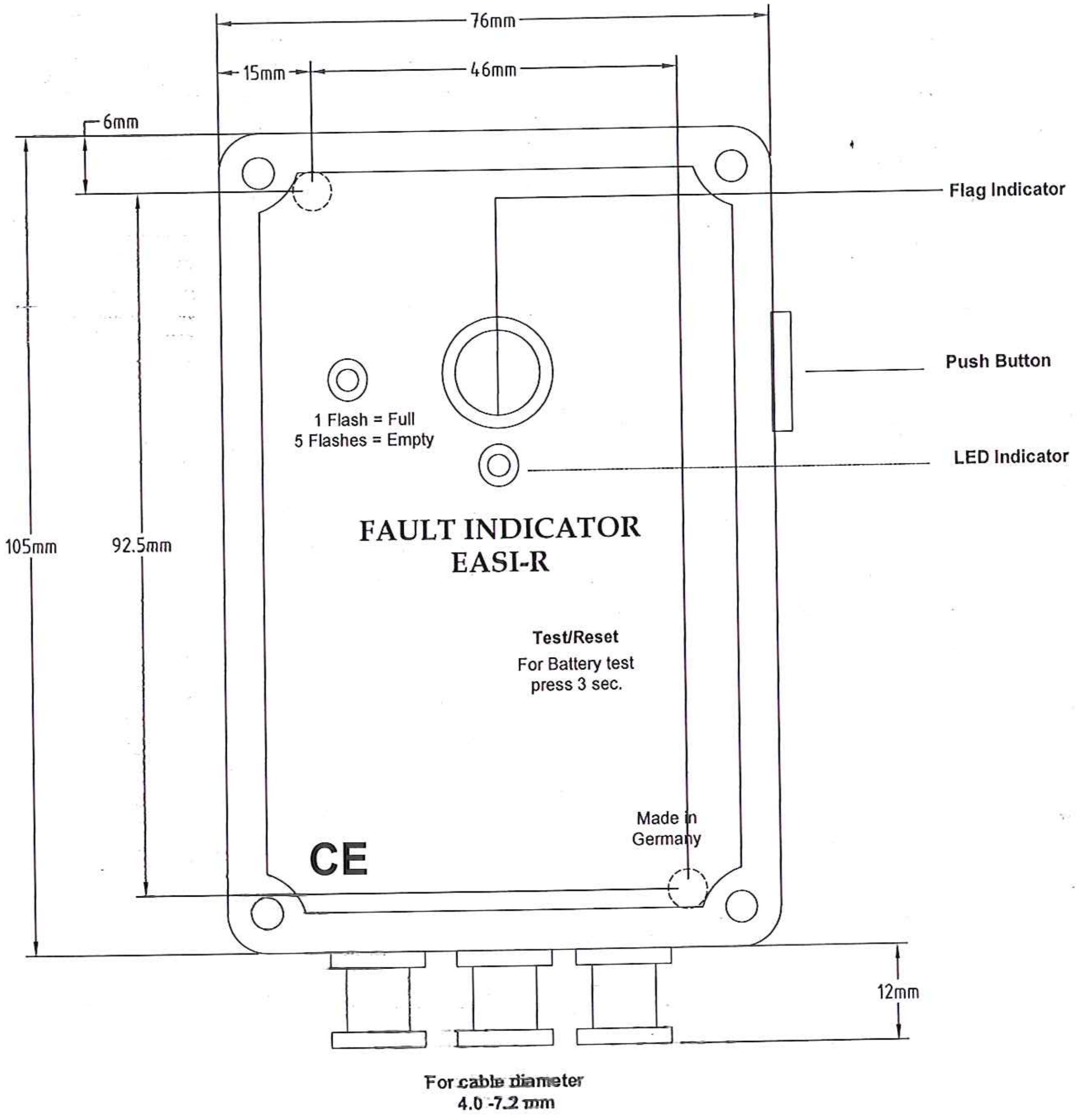
Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No : CSU-SCA-EFI
Sheet 2/9

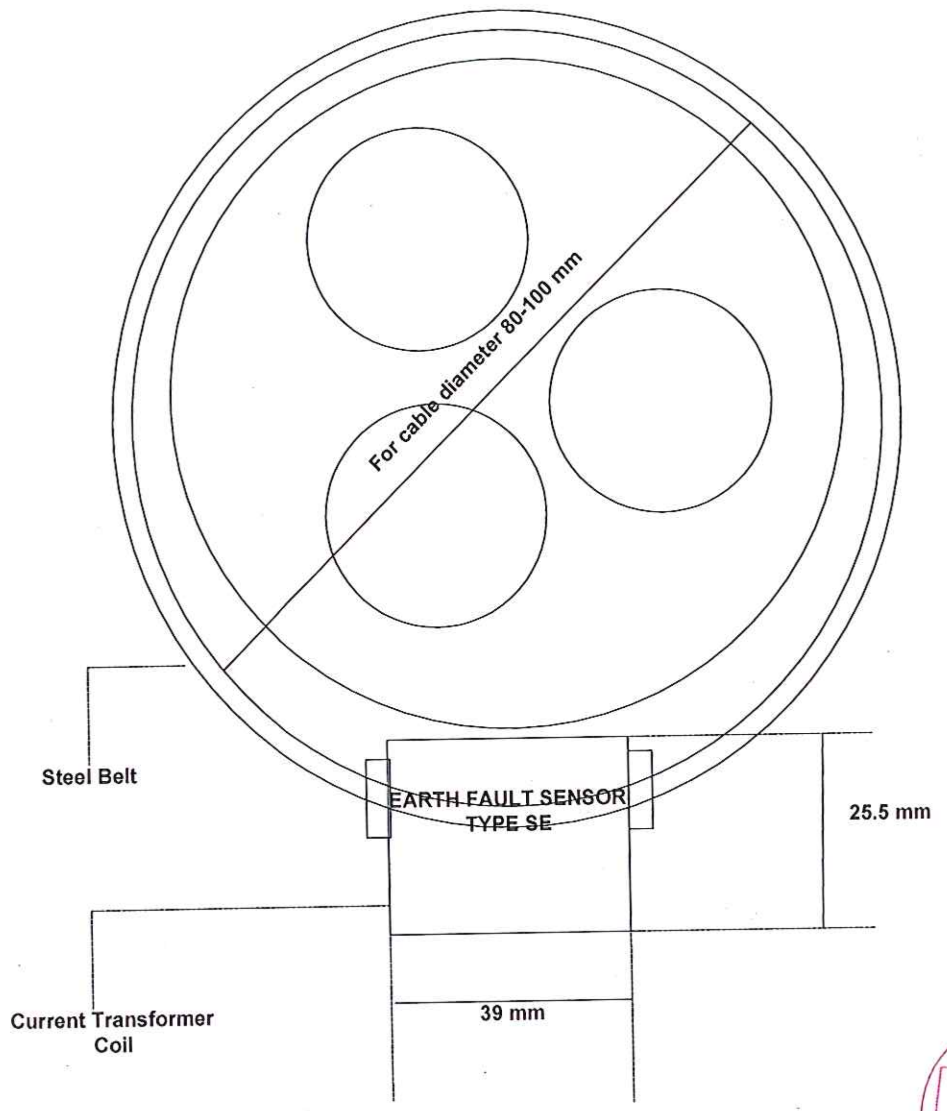
IM Code : -
Tender No. : -

Dept.	Year	Scale	Units
ENG	2017	-	mm

FORM NO: QA-F-01-12F


76mm
15mm
46mm
6mm
105mm
92.5mm
12mm
Flag Indicator
Push Button
LED Indicator
1 Flash = Full
5 Flashes = Empty
FAULT INDICATOR
EASI-R
Test/Reset
For Battery test
press 3 sec.
CE
Made in
Germany
For cable diameter
4.0 -7.2 mm

Earth Fault
Indicator Unit


For cable diameter 80-100 mm
Steel Belt
EARTH FAULT SENSOR
TYPE SE
25.5 mm
39 mm
Current Transformer
Coil

CT Sensor

AMSDS
20 OCT 2017
TNB

Rev.	Reason	X	Y	Day	Month	Year
00	INITIAL RELEASE			20	OCT	2017

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : DIMENSION EMG EFI EASI-R DIMENSION

MATERIAL
Type : -
Code : -

COATING
Type : -
Code : -

GENERAL TOLERANCE
X.X ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABDUL MANAN

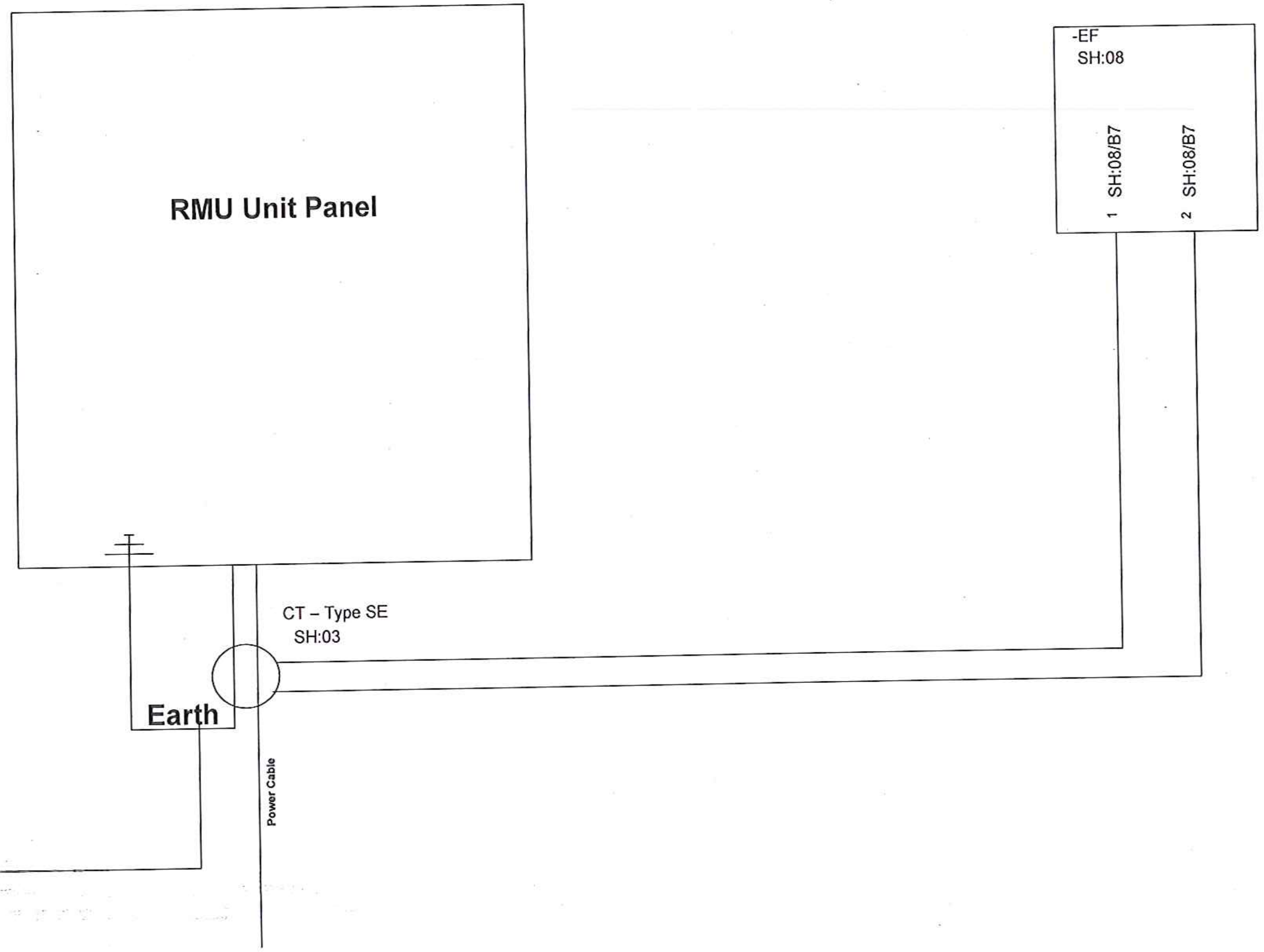
Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No : CSU-SCA-EFI
Sheet 3/9

IM Code : -
Tender No. : -

Dept.	Year	Scale	Units
ENG	2017	-	mm

FORM NO: QA-F-01-12F


RMU Unit Panel
-EF
SH:08
1 SH:08/B7
2 SH:08/B7
CT – Type SE
SH:03
Earth
Power Cable

AMSDS
20 OCT 2017
TNB

Rev.	Reason	X	Y	Day	Month	Year
00	INITIAL RELEASE			20	OCT	2017

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : CT CONNECTION

MATERIAL
Type : -
Code : -

COATING
Type : -
Code : -

GENERAL TOLERANCE
X.X ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABD. MANAN

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

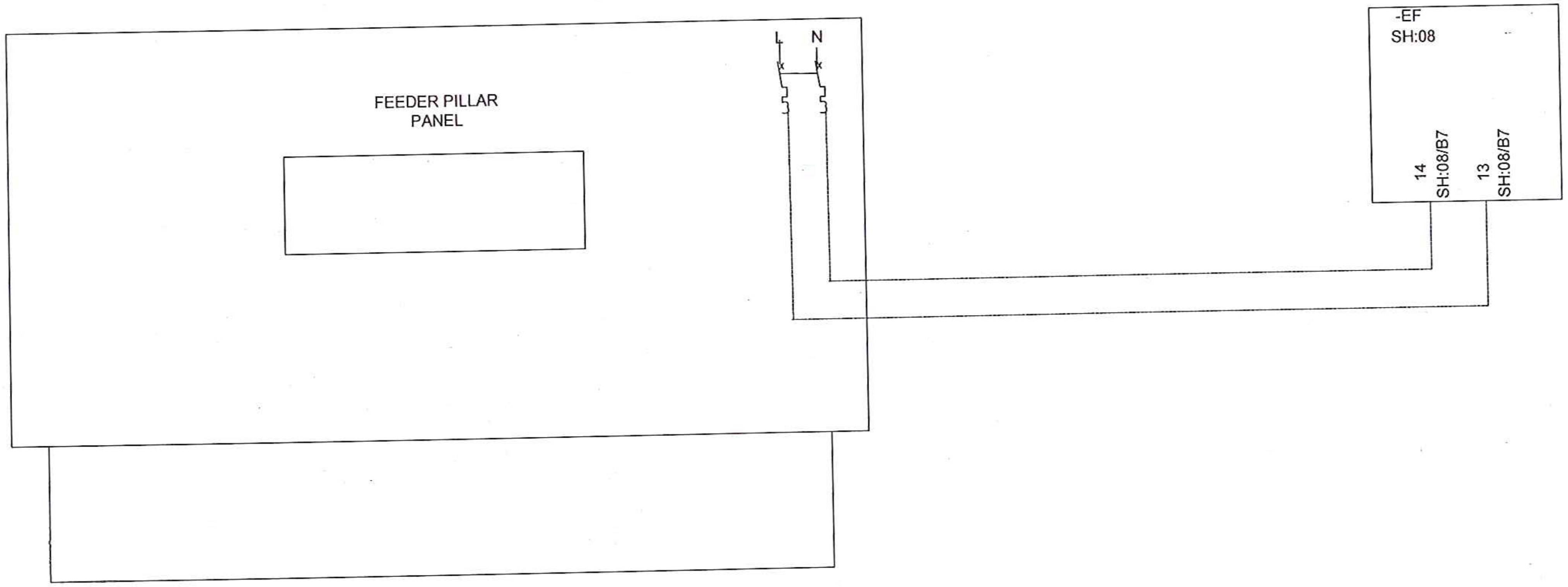
Dwg No : CSU-SCA-EFI

Sheet 4/9

IM Code : -
Tender No. : -

Dept.	Year	Scale	Units
ENG	2017	-	mm

FORM NO: QA-F-01-12F



Note : - Suppression of inrush current with a length of up to 3 Sec only available with connected 240 Vac that fails during fault, connector 13-14, selectable by dip switch.

AMSDS
20 OCT 2017
TNB

00	INITIAL RELEASE			20	OCT	2017
Rev.	Reason	X	Y	Day	Month	Year

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : VOLTAGE INPUT CONNECTION (OPTION)

MATERIAL	Type : - Code : -	COATING	Type : - Code : -

GENERAL TOLERANCE
X.X ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABD. MANAN

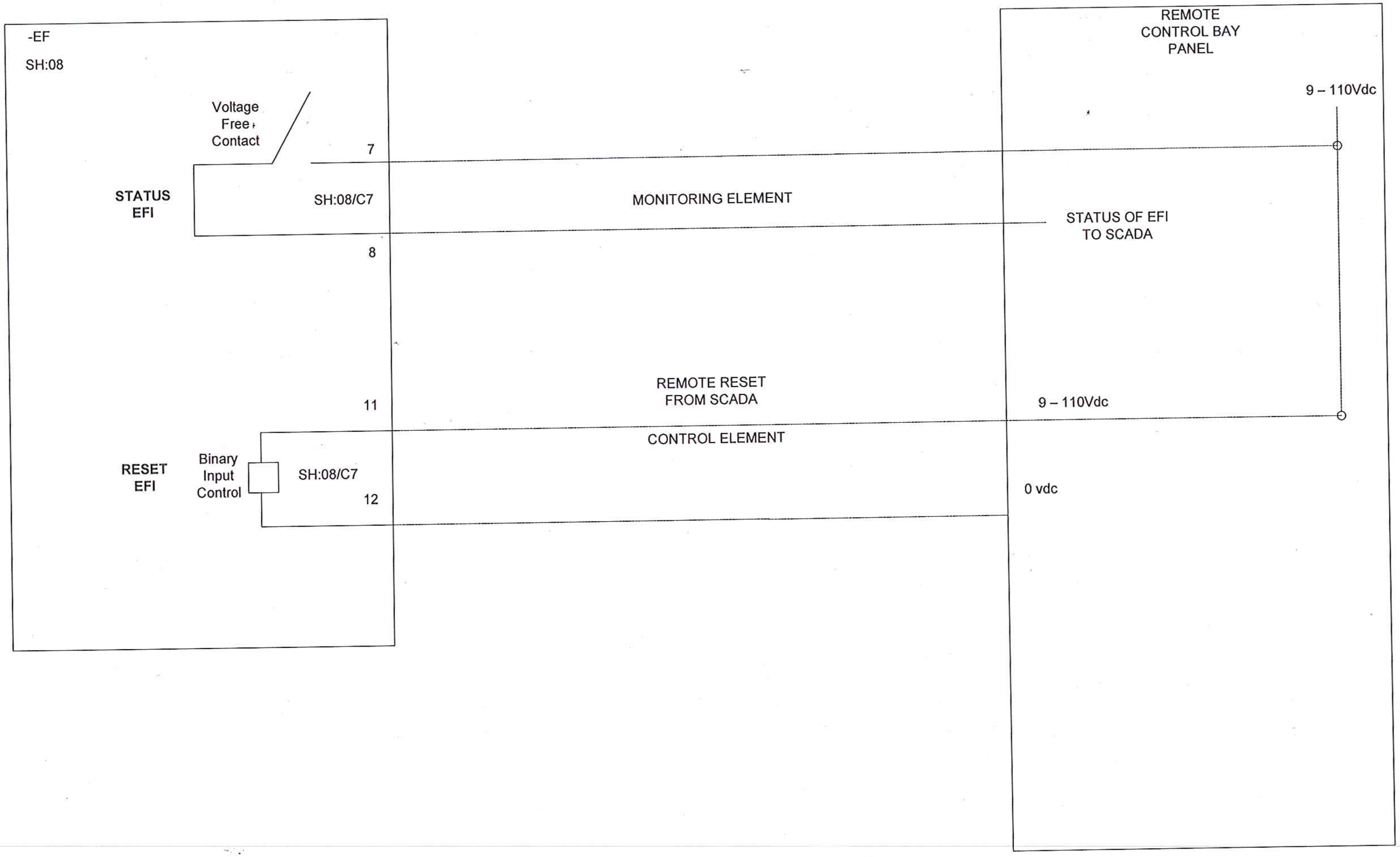
Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No : CSU-SCA-EFI
Sheet 5/9

IM Code : -
Tender No. : -

Dept.	Year	Scale	Units
ENG	2017	-	mm

FORM NO: QA-F-01-12F


-EF
SH:08
Voltage Free Contact
7
STATUS EFI
SH:08/C7
8
MONITORING ELEMENT
REMOTE CONTROL BAY PANEL
9 – 110Vdc
STATUS OF EFI TO SCADA
REMOTE RESET FROM SCADA
11
9 – 110Vdc
CONTROL ELEMENT
RESET EFI
Binary Input Control
SH:08/C7
12
0 vdc

AMSDS
20 OCT 2017

Rev.	Reason	X	Y	Day	Month	Year
00	INITIAL RELEASE			20	OCT	2017

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : SCADA CONNECTION 1 (OPTION 1)

MATERIAL	Type : - Code : -	COATING	Type : - Code : -

GENERAL TOLERANCE
X.X ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABDUL MANAN

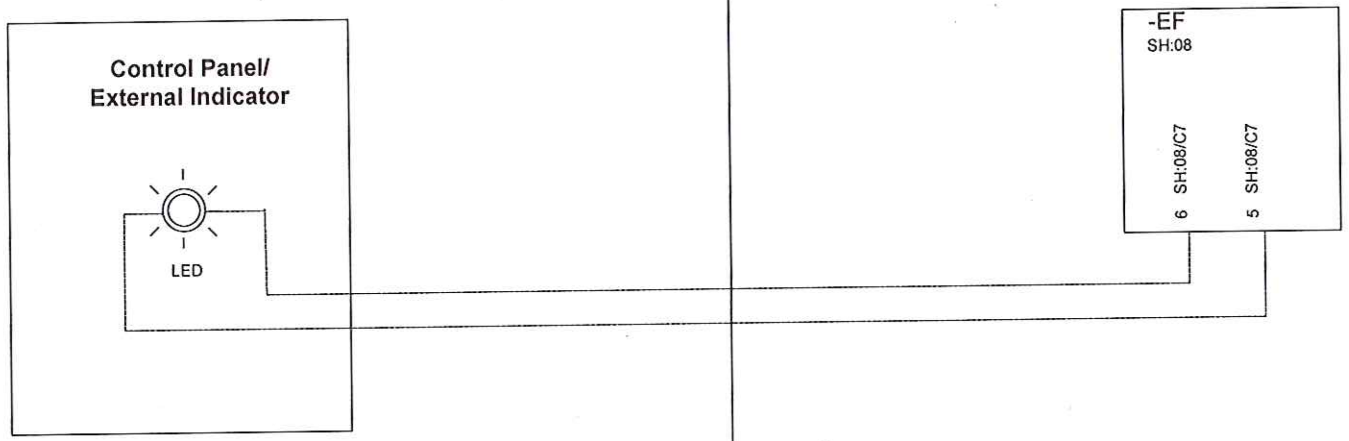
Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

IM Code : -
Tender No. : -

Dwg No : CSU-SCA-EFI
Sheet 6/9

Dept.	Year	Scale	Units	Sheet
ENG	2017	-	mm	1/1

FORM NO: QA-F-01-12F


Control Panel/
External Indicator
LED
-EF
SH:08
6 SH:08/C7
5 SH:08/C7

00	INITIAL RELEASE			20	OCT	2017
Rev.	Reason	X	Y	Day	Month	Year

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : EXTERNAL LED (OPTION)

MATERIAL	Type : - Code : -	COATING	Type : - Code : -

GENERAL TOLERANCE
X.X ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABD. MANAN

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No : CSU-SCA-EFI

Sheet 7/9

IM Code : -
Tender No. : -

Dept.	Year	Scale	Units
ENG	2017	-	mm

FORM NO: QA-F-01-12F

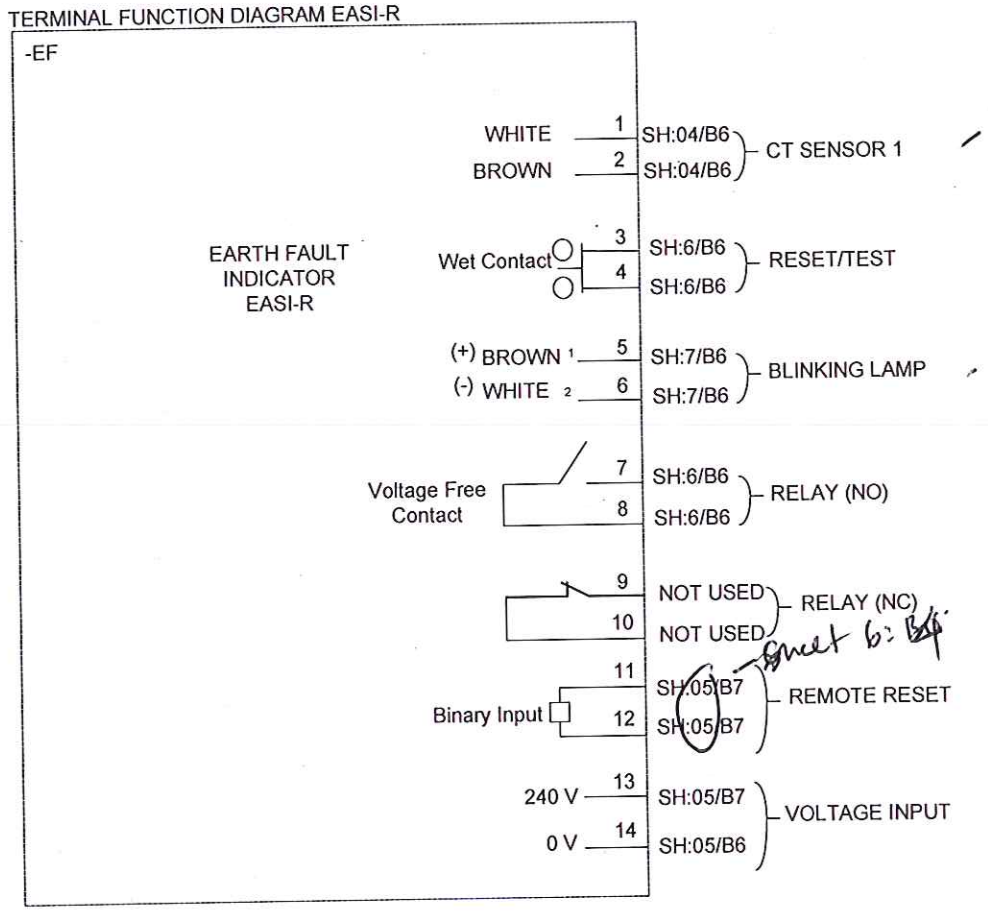
DISTRIBUTION NETWORK
AMSDS
20 OCT 2017
TNB

DEVICE FUNCTION : TERMINAL FUNCTION DRAWING EFI EASI-R
DEVICE : -EFI
MAKE : EMG
TYPE : EASI-R

TECHNICAL DATA

Earth Fault Current (phase to ground)	Adjustable : 40/80/120/160/240 A (tolerance +/- 10%)
Response Delay	Adjustable : 50/100/150/200 ms
Display Unit	Suitable for Surface Installation
Response Delay	Adjustable : 50/100/150/200 ms
Indication of Earth Fault	A) Red LED B) Flag Indication
Indication of battery status	Yellow LED
Reset of the Indicator	A) MANUAL push button B) SCADA connection for a potential free remote reset C) TIME : 4/8/16/32 (+/- 1%) hours after fault D) AUTO RESET , self acting after recovering net current (selectable by DIP switch) E) AUTO RESET , self acting after recovering 230Vac (Selectable by DIP switch)
On site function test	A) the button has to be pressed for 1 sec (by push button) B) the button has to be pressed for 3 sec (by push button)
Remote function test	By closing potential-free test/reset contact for 5-10 sec (connector 3 and 4)
Inrush current restraint	Suppression of inrush currents with a length of up to 3 sec (only available with connected 230V AC that fail during fault, connector 13-14 Selectable by DIP switch)
Dimension (Display Unit)	(W x H X D) 76mm x 117mm x 43mm
Protection Class (Display Unit)	IP 65
Protection Class (Sensors)	IP 67
EMI	According to IEC 1000-4-2, IEC 1000-4-3
Internal Type Test	According to IEEE 495-2007
Operation Temperature Range	-25 C to + 70 C
Maximum Humidity	95%
Power Supply	Lithium Battery (LiSOCl2) type AA / 3.6V / 2600mAh
SCADA Contact	1 x NO and 1 x NC Permanent / wipe contact (100ms) (contact type can be selected on site by a dip switch) Max. 230 – 240 Vac / max. 2A /max. 30 W Relay material : gold plated silver against palladium silver
Earth Fault sensor (CT)	1 earth fault sensor type SE (current transformer for a three-core cable or three single core cable) Diameter : 80-100mm Connection cable length : 16m (copper cable)

TERMINAL FUNCTION DIAGRAM EASI-R


-EF
EARTH FAULT INDICATOR EASI-R
WHITE
BROWN
1 SH:04/B6
2 SH:04/B6
CT SENSOR 1
Wet Contact
3 SH:6/B6
4 SH:6/B6
RESET/TEST
(+) BROWN 1
(-) WHITE 2
5 SH:7/B6
6 SH:7/B6
BLINKING LAMP
Voltage Free Contact
7 SH:6/B6
8 SH:6/B6
RELAY (NO)
9 NOT USED
10 NOT USED
RELAY (NC)
Sheet 6: B4
Binary Input
11 SH:05/B7
12 SH:05/B7
REMOTE RESET
240 V
0 V
13 SH:05/B7
14 SH:05/B6
VOLTAGE INPUT

AMSDS
20 OCT 2017
TNB

MATERIAL	Type : - Code : -	COATING	Type : - Code : -

GENERAL TOLERANCE
XX ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

Rev.	Reason	X	Y	Day	Month	Year
00	INITIAL RELEASE			20	OCT	2017

Title : COMPACT SUBSTATION UNIT SCADA EQUIPPED 500kVA & 1000kVA
Description : TERMINAL FUNCTION DIAGRAM

INDKOM
ENGINEERING SDN. BHD.

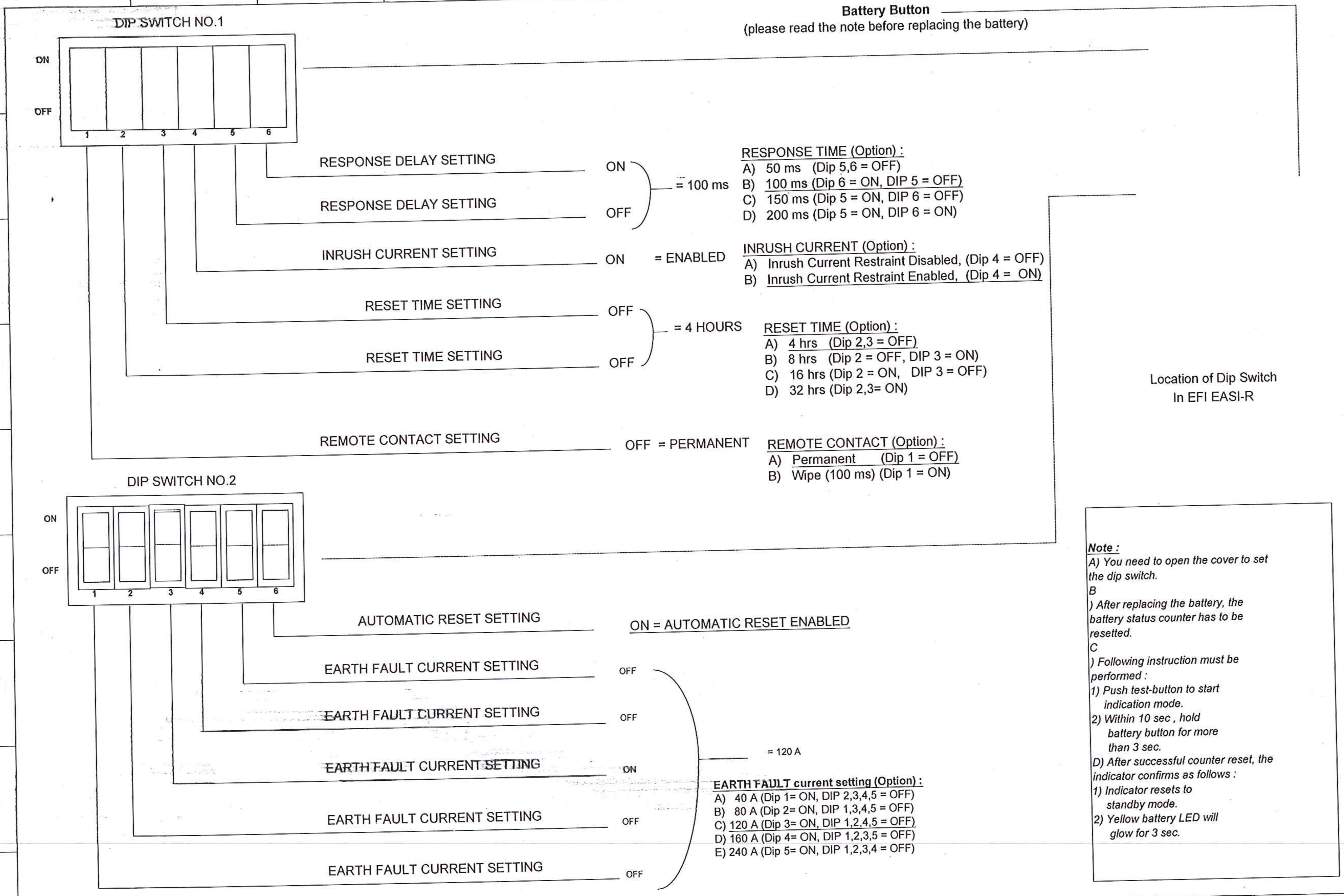
Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABD. MANAN

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION
Dwg No : CSU-SCA-EFI
Sheet 8 / 9

IM Code : -
Tender No. : -

Dept.	Year	Scale	Units
ENG	2017	-	mm

FORM NO: QA-F-01-12F


DIP SWITCH NO.1
ON
OFF
1
2
3
4
5
6
Battery Button
(please read the note before replacing the battery)
RESPONSE DELAY SETTING
ON
RESPONSE DELAY SETTING
OFF
= 100 ms
INRUSH CURRENT SETTING
ON
= ENABLED
RESET TIME SETTING
OFF
RESET TIME SETTING
OFF
= 4 HOURS
REMOTE CONTACT SETTING
OFF = PERMANENT
Location of Dip Switch
In EFI EASI-R
DIP SWITCH NO.2
ON
OFF
1
2
3
4
5
6
AUTOMATIC RESET SETTING
ON = AUTOMATIC RESET ENABLED
EARTH FAULT CURRENT SETTING
OFF
EARTH FAULT CURRENT SETTING
OFF
EARTH FAULT CURRENT SETTING
ON
EARTH FAULT CURRENT SETTING
OFF
EARTH FAULT CURRENT SETTING
OFF
= 120 A

RESPONSE TIME (Option) :
A) 50 ms (Dip 5,6 = OFF)
B) 100 ms (Dip 6 = ON, DIP 5 = OFF)
C) 150 ms (Dip 5 = ON, DIP 6 = OFF)
D) 200 ms (Dip 5 = ON, DIP 6 = ON)

INRUSH CURRENT (Option) :
A) Inrush Current Restraint Disabled, (Dip 4 = OFF)
B) Inrush Current Restraint Enabled, (Dip 4 = ON)

RESET TIME (Option) :
A) 4 hrs (Dip 2,3 = OFF)
B) 8 hrs (Dip 2 = OFF, DIP 3 = ON)
C) 16 hrs (Dip 2 = ON, DIP 3 = OFF)
D) 32 hrs (Dip 2,3= ON)

REMOTE CONTACT (Option) :
A) Permanent (Dip 1 = OFF)
B) Wipe (100 ms) (Dip 1 = ON)

EARTH FAULT current setting (Option) :
A) 40 A (Dip 1= ON, DIP 2,3,4,5 = OFF)
B) 80 A (Dip 2= ON, DIP 1,3,4,5 = OFF)
C) 120 A (Dip 3= ON, DIP 1,2,4,5 = OFF)
D) 160 A (Dip 4= ON, DIP 1,2,3,5 = OFF)
E) 240 A (Dip 5= ON, DIP 1,2,3,4 = OFF)

Note :
A) You need to open the cover to set the dip switch.
B
) After replacing the battery, the battery status counter has to be resetted.
C
) Following instruction must be performed :
1) Push test-button to start indication mode.
2) Within 10 sec , hold battery button for more than 3 sec.
D) After successful counter reset, the indicator confirms as follows :
1) Indicator resets to standby mode.
2) Yellow battery LED will glow for 3 sec.

AMSDS
20 OCT 2017
TNB

MATERIAL
Type : -
Code : -
COATING
Type : -
Code : -
GENERAL TOLERANCE
X.X ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED
PROJECTION

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA
Description : DIP SWITCH SETTING

INDKOM
ENGINEERING SDN. BHD.

Approved by : EN. JAMAL
Checked by : EN. ALI
Drawn by : ABD. MANAN

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION
Dwg No : CSU-SCA-EFI
Sheet 9/9
IM Code : -
Tender No. : -
Dept. ENG
Year 2017
Scale -
Units mm

Rev.	Reason	X	Y	Day	Month	Year
00	INITIAL RELEASE			20	OCT	2017

FORM NO: QA-F-01-12F